

SOLUTION OF QUESTION 1: SAMPLING IN SGD

See Winter 2024 Moed B question 3

SOLUTION OF QUESTION 2: SAMPLING IN SGD

1. Using the hint.

$$\mathcal{L}(\mathbf{w} - \eta \hat{\mathbf{g}}) \leq U(\eta) \triangleq \mathcal{L}(\mathbf{w}) - \eta \mathbf{g}^\top \hat{\mathbf{g}} + \frac{1}{2} \eta^2 \beta \|\hat{\mathbf{g}}\|^2$$

$$\mathbb{E} \mathcal{L}(\mathbf{w} - \eta \hat{\mathbf{g}}) \leq \mathbb{E} U(\eta) = \mathcal{L}(\mathbf{w}) - \eta \mathbf{g}^\top \mathbb{E} \hat{\mathbf{g}} + \frac{1}{2} \eta^2 \beta \mathbb{E} [\|\hat{\mathbf{g}}\|^2]$$

By differentiating the bound we find the optimal η_t :

$$-\mathbf{g}^\top \mathbb{E} \hat{\mathbf{g}} + \eta \beta \mathbb{E} [\|\hat{\mathbf{g}}\|^2] = 0$$

$$\eta = \frac{\mathbf{g}^\top \mathbb{E} \hat{\mathbf{g}}}{\beta \mathbb{E} [\|\hat{\mathbf{g}}\|^2]}$$

which is a valid step size since $\mathbf{g}^\top \mathbb{E} \hat{\mathbf{g}} > 0$ and $\mathbb{E} [\|\hat{\mathbf{g}}\|^2] \neq 0$. Plugging it into the bound, we get

$$\mathbb{E} \mathcal{L}(\mathbf{w} - \eta \hat{\mathbf{g}}) \leq \mathcal{L}(\mathbf{w}) - \frac{(\mathbf{g}^\top \mathbb{E} \hat{\mathbf{g}})^2}{2\beta \mathbb{E} [\|\hat{\mathbf{g}}\|^2]}.$$

2. Note that

$$\begin{aligned} \mathbb{E} [\|\hat{\mathbf{g}}\|^2] &= \mathbb{E} [\|\hat{\mathbf{g}} - \mathbb{E} \hat{\mathbf{g}} + \mathbb{E} \hat{\mathbf{g}}\|^2] = \mathbb{E} [\|\hat{\mathbf{g}} - \mathbb{E} \hat{\mathbf{g}}\|^2 + \|\mathbb{E} \hat{\mathbf{g}}\|^2 + 2\mathbb{E} \hat{\mathbf{g}}^\top (\hat{\mathbf{g}} - \mathbb{E} \hat{\mathbf{g}})] \\ &= \mathbb{E} [\|\hat{\mathbf{g}} - \mathbb{E} \hat{\mathbf{g}}\|^2] + \|\mathbb{E} \hat{\mathbf{g}}\|^2 + 2\mathbb{E} \hat{\mathbf{g}}^\top (\mathbb{E} \hat{\mathbf{g}} - \mathbb{E} \hat{\mathbf{g}}) = \mathbb{E} [\|\hat{\mathbf{g}} - \mathbb{E} \hat{\mathbf{g}}\|^2] + \|\mathbb{E} \hat{\mathbf{g}}\|^2 \\ &= \text{Var}(\hat{\mathbf{g}}) + \|\mathbb{E} \hat{\mathbf{g}}\|^2 \end{aligned}$$

so we can write

$$\mathbb{E} \mathcal{L}(\mathbf{w} - \eta \hat{\mathbf{g}}) \leq \mathcal{L}(\mathbf{w}) - \frac{(\mathbf{g}^\top \mathbb{E} \hat{\mathbf{g}})^2}{2\beta (\text{Var}(\hat{\mathbf{g}}) + \|\mathbb{E} \hat{\mathbf{g}}\|^2)}.$$

From the hint, the numerator is maximized when $\mathbb{E} \hat{\mathbf{g}} = c\mathbf{g}$, with $c > 0$ since $\mathbf{g}^\top \mathbb{E} \hat{\mathbf{g}} > 0$ and $c = 1$ since $\|\mathbb{E} \hat{\mathbf{g}}\| = \|\mathbf{g}\|$. Therefore, we need an unbiased estimator, i.e. $\mathbb{E} \hat{\mathbf{g}} = \mathbf{g}$. Also, the denominator is minimized when $\text{Var}(\hat{\mathbf{g}})$ is minimized.

3. In this case

$$\mathbb{E} [\hat{\mathbf{g}}] = \mathbb{E} \left[\frac{1}{N} \sum_{n=1}^N c_n I_n \right] = \frac{1}{N} \sum_{n=1}^N c_n \mathbb{E} [I_n] = \frac{1}{N} \sum_{n=1}^N c_n p_n \stackrel{c_n=1/p_n}{=} \frac{1}{N} \sum_{n=1}^N 1 = \mathbf{g}$$

where in the last equality we chose $c_n = 1/p_n$.

4. In this case

$$\begin{aligned}
\text{Var}(\hat{\mathbf{g}}) &= \mathbb{E} \|\hat{\mathbf{g}} - \mathbb{E}\hat{\mathbf{g}}\|^2 \\
&= \mathbb{E} \left\| \frac{1}{N} \sum_{n=1}^N \frac{1}{p_n} I_n \mathbf{g}_n - \frac{1}{N} \sum_{n=1}^N \mathbf{g}_n \right\|^2 \\
&= \mathbb{E} \left\| \frac{1}{N} \sum_{n=1}^N \left(\frac{1}{p_n} I_n - 1 \right) \mathbf{g}_n \right\|^2 \\
&= \frac{1}{N^2} \mathbb{E} \left[\sum_{n=1}^N \sum_{m=1}^N \left(\frac{1}{p_n} I_n - 1 \right) \left(\frac{1}{p_m} I_m - 1 \right) \mathbf{g}_m^\top \mathbf{g}_n \right] \\
&= \frac{1}{N^2} \sum_{n=1}^N \sum_{m=1}^N \mathbb{E} \left[\left(\frac{1}{p_n} I_n - 1 \right) \left(\frac{1}{p_m} I_m - 1 \right) \right] \mathbf{g}_m^\top \mathbf{g}_n \\
&= \frac{1}{N^2} \sum_{n=1}^N \sum_{m=1}^N \mathbb{E} \left[\frac{1}{p_n p_m} I_m I_n - \frac{1}{p_m} I_m - \frac{1}{p_n} I_n + 1 \right] \mathbf{g}_m^\top \mathbf{g}_n \\
&= \frac{1}{N^2} \sum_{n=1}^N \sum_{m=1}^N \left[\frac{1}{p_n p_m} ((p_n - p_n^2) \delta_{mn} + p_n p_m) - 1 \right] \mathbf{g}_m^\top \mathbf{g}_n \\
&= \frac{1}{N^2} \sum_{n=1}^N \sum_{m=1}^N \left[\frac{1}{p_n p_m} (p_n - p_n^2) \delta_{mn} \right] \mathbf{g}_m^\top \mathbf{g}_n \\
&= \frac{1}{N^2} \sum_{n=1}^N \left[\frac{1}{p_n^2} (p_n - p_n^2) \right] \|\mathbf{g}_n\|^2 \\
&= \frac{1}{N^2} \sum_{n=1}^N \frac{1 - p_n}{p_n} \|\mathbf{g}_n\|^2
\end{aligned}$$

5. As hinted, we use Lagrange multipliers to enforce the constraint that $\sum_{n=1}^N p_n = 1$. The Lagrangian is

$$\mathcal{L} = \sum_{n=1}^N \left(\frac{1}{p_n} - 1 \right) \|\mathbf{g}_n\|^2 + \lambda \left(\sum_{n=1}^N p_n - 1 \right)$$

it's stationary point condition is

$$\frac{\partial}{\partial p_k} \mathcal{L} = -\frac{1}{p_k} \|\mathbf{g}_k\|^2 + \lambda = 0$$

which requires

$$p_k = \lambda^{-1} \|\mathbf{g}_k\|^2.$$

To make sure that p_k are normalized, i.e. $\sum_{n=1}^N p_n = 1$, we set $\lambda = \sum_{n=1}^N \|\mathbf{g}_n\|^2$ and obtain

$$p_k = \frac{\|\mathbf{g}_k\|^2}{\sum_{n=1}^N \|\mathbf{g}_n\|^2}.$$

Note that this solution also fulfills the standard constraints on probability: $p_k \in [0, 1]$.

6. Since need to calculate all the per-sample gradients before taking a single step, this method has the same computational price as GD, and it much more expensive than standard SGD.

SOLUTION OF QUESTION 3: MIXTURE OF EXPERTS

1. This can be useful for inference since, due to the TopK operation, in each we only need to use the TopK experts instead of M . This operation is useful only if the computational complexity of doing the mixture, using the $g_i(\text{TopK}(\sigma(\mathbf{W}\mathbf{x})))$ selection operation is significantly cheaper than using each expert $f_{\theta_i}(\mathbf{x})$. In other words, MoE are useful when the experts are sufficiently large in comparison to the selection operation. Note that this model is more flexible (i.e., expressive) than just having a fixed set of K experts, since we can use a different set of K expert for each input $\mathbf{x}$.

2. We have a discontinuity whenever the location of the K largest element in $\mathbf{v}$ changes. Otherwise, the function is continuous. For example, for $K = 1$, we have

$$\text{Top1}((x + \epsilon, x - \epsilon)) = \begin{cases} (x + \epsilon, 0) & , \text{ if } \epsilon > 0 \\ (0, x - \epsilon) & , \text{ if } \epsilon < 0 \end{cases}$$

so we see the value of the each component jumps by x (or $-x$) when $\epsilon = 0$.

3. For $K = 1$, if we denote $\mathbf{z} = \mathbf{W}\mathbf{x}$, then

$$g_i(\text{TopK}(\sigma(\mathbf{z}))) = \begin{cases} 1 & , \text{ if } z_i = \max_j z_j \\ 0 & , \text{ else} \end{cases} ,$$

i.e. a piece-wise constant function. For example, if there is only one maximum (the generic case), we get a one-hot encoding of the index of the maximum in $\mathbf{z}$. In this case, the derivative

$$\nabla_{W_{i,j}} \text{MoE}(\mathbf{x}) = f_{\theta_i}(\mathbf{x}) \nabla_{W_{i,j}} g_i$$

is a zero almost everywhere (and infinity in the boundaries where the index of the maximum changes). In this case, we cannot optimize $\mathbf{W}$ with the standard gradient based methods, such as SGD (if we do not use any approximation for the gradient, such as the STE we learned in class).

4. We have

$$(3.1) \quad \nabla_{\theta_i} \text{MoE}(\mathbf{x}) = g_i \nabla_{\theta_i} f_{\theta_i}(\mathbf{x}) ,$$

And, denoting $\mathbf{v} = \text{TopK}(\sigma(\mathbf{W}\mathbf{x}))$, $\mathbf{w}_m$ as the m 'th row in $\mathbf{W}$ and

$$I(A) = \begin{cases} 1 & , \text{ if } A \\ 0 & , \text{ else} \end{cases}$$

we get

$$\nabla_{W_{i,j}} \text{MoE}(\mathbf{x})$$

(3.2)

$$= \sum_{m=1}^M f_{\theta_m}(\mathbf{x}) \nabla_{W_{i,j}} g_m$$

$$= \sum_{m=1}^M I(v_m \neq 0) f_{\theta_m}(\mathbf{x}) \left(\frac{\nabla_{W_{i,j}} v_m}{\sum_{j'=1}^M I(v_{j'} \neq 0) v_{j'}} - \frac{\nabla_{W_{i,j}} v_i}{\left(\sum_{j'=1}^M I(v_{j'} \neq 0) v_{j'}\right)^2} I(v_i \neq 0) \right)$$

(3.3)

$$= \sum_{m=1}^M I(v_m \neq 0) f_{\theta_m}(\mathbf{x}) \left(\frac{\sigma'(\mathbf{w}_m^\top \mathbf{x}) x_j}{\sum_{j'=1}^M I(v_{j'} \neq 0) v_{j'}} I(m=i) - \frac{\sigma'(\mathbf{w}_j^\top \mathbf{x}) x_j}{\left(\sum_{j'=1}^M I(v_{j'} \neq 0) v_{j'}\right)^2} I(v_i \neq 0) \right)$$

5. Note that we need to calculate the experts gradient $\nabla_{\theta_i} \text{MoE}(\mathbf{x})$ in equation 3.1 only for i indices in the TopK, since all the other gradients are zero. Likewise, we see from the expression for $\nabla_{W_{i,j}} \text{MoE}(\mathbf{x})$, the gradient of the gating parameters, that only the topK components are used in the expression (for which $v_i \neq 0$). Therefore, we also get a significant saving during training.

6. If the weights $\mathbf{W}$ become large (a common case in practice), then if $\sigma(x)$ was replaced with the common e^x , then the maximal exponent will dominate and we will get

$$g_i(\text{TopK}(\sigma(\mathbf{z}))) \rightarrow \begin{cases} 1 & , \text{ if } z_i = \max_j z_j \\ 0 & , \text{ else} \end{cases}$$

i.e., the Top1 selector. This has less expressive power than TopK selector for $K > 1$ since it only uses only one expert (no matter what is K).